



RS003 Week 16 — English Worksheet

Topic: Polish — Missiles, Sprites, High Score, and Show & Tell · **Course:** Pygame Space Shooter ·

Time: about 45 minutes

This is the final week. You add the **polish** that makes a game feel finished: a second weapon (**missiles** with the M key) that hits harder, a **high score** that remembers your best, a restart key, and best of all you swap your drawn shapes for real **PNG sprites** loaded from image files. Then you present your finished game.

Words to know: **missile**, **sprite**, **image.load**, **high score**, **restart**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
ship_img = pygame.image.load("ship.png").convert_alpha()
enemy_img = pygame.image.load("enemy.png").convert_alpha()
screen.blit(ship_img, (ship_x - ship_img.get_width() // 2, ship_y))
```

These lines load picture files and put them on screen. What replaces your drawn rectangles and triangles?


```
if event.key == pygame.K_m:
    missiles.append([ship_x, ship_y - 20])
```

Pressing M fires a missile instead of a bullet. Why might a game want a second, stronger weapon?


```
high_score = 0
if event.key == pygame.K_r and game_over:
    if score > high_score:
        high_score = score
    score = 0
    game_over = False
```

Pressing R restarts the game. When does the high score update?

2 · Run It

 **Type the example, run it, and fire both weapons**

💡 For the sprite version below you need two small picture files saved next to your code: `ship.png` and `enemy.png` . If you finished the **AI Art** extension, use those. No images yet? Use the drawn shapes from week 15 and add `ship.png` / `enemy.png` once you have them.

```

import pygame, random
pygame.init()
WIDTH, HEIGHT = 800, 600
screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Space Shooter")
font = pygame.font.SysFont(None, 40)
big_font = pygame.font.SysFont(None, 72)
clock = pygame.time.Clock()

SPACE = (10, 12, 40); WHITE = (255, 255, 255)
RED = (230, 60, 60); YELLOW = (255, 210, 40); GREY = (120, 120, 140)

ship_img = pygame.image.load("ship.png").convert_alpha()
enemy_img = pygame.image.load("enemy.png").convert_alpha()

ship_x, ship_y = WIDTH // 2, HEIGHT - 100
SPEED = 5
bullets = []          # each: [x, y]
missiles = []         # each: [x, y]
enemies = []          # each: {"x","y","type","points"}
effects = []           # each: [x, y, timer]
score = 0
high_score = 0
spawn_timer = 0
game_over = False
start_time = pygame.time.get_ticks()
GAME_TIME = 30000

running = True
while running:
    elapsed = pygame.time.get_ticks() - start_time
    time_left = max(0, (GAME_TIME - elapsed) // 1000)
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
        if event.type == pygame.KEYDOWN:
            if event.key == pygame.K_SPACE and not game_over:
                bullets.append([ship_x, ship_y - 20])
            if event.key == pygame.K_m and not game_over:
                missiles.append([ship_x, ship_y - 20])
            if event.key == pygame.K_r and game_over:
                if score > high_score:
                    high_score = score
                score = 0
                enemies.clear()

```

```

        start_time = pygame.time.get_ticks()
        game_over = False

if time_left == 0:
    game_over = True

if not game_over:
    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]: ship_x -= SPEED
    if keys[pygame.K_RIGHT]: ship_x += SPEED
    if keys[pygame.K_UP]: ship_y -= SPEED
    if keys[pygame.K_DOWN]: ship_y += SPEED
    ship_x = max(20, min(WIDTH - 20, ship_x))
    ship_y = max(20, min(HEIGHT - 60, ship_y))

    spawn_timer += 1
    if spawn_timer >= 40:
        spawn_timer = 0
        kind = random.choice(["alien", "meteor"])
        points = 1 if kind == "alien" else 3
        enemies.append({"x": random.randint(40, WIDTH - 40), "y": -20, "type":
kind, "points": points})

    for e in enemies: e["y"] += 2
    for b in bullets: b[1] -= 8
    for mi in missiles: mi[1] -= 4

    for e in enemies[:]:
        for b in bullets[:]:
            if abs(b[0] - e["x"]) < 22 and abs(b[1] - e["y"]) < 22:
                score += e["points"]
                effects.append([e["x"], e["y"], 8])
                if e in enemies: enemies.remove(e)
                if b in bullets: bullets.remove(b)
        for mi in missiles[:]:
            if abs(mi[0] - e["x"]) < 30 and abs(mi[1] - e["y"]) < 30:
                score += e["points"] * 2
                effects.append([e["x"], e["y"], 12])
                if e in enemies: enemies.remove(e)
                if mi in missiles: missiles.remove(mi)

    bullets = [b for b in bullets if b[1] > -10]
    missiles = [m for m in missiles if m[1] > -10]
    enemies = [e for e in enemies if e["y"] < HEIGHT + 20]

```

```

    screen.fill(SPACE)
    for b in bullets:  pygame.draw.rect(screen, YELLOW, (b[0] - 2, b[1] - 8, 4,
12))
    for mi in missiles: pygame.draw.rect(screen, WHITE, (mi[0] - 4, mi[1] - 12, 8,
18))
    for e in enemies:  screen.blit(enemy_img, (e["x"] - enemy_img.get_width() //
2, e["y"] - enemy_img.get_height() // 2))
    for fx in effects: pygame.draw.circle(screen, YELLOW, (fx[0], fx[1]), fx[2])
    effects = [[x, y, t - 1] for x, y, t in effects if t > 1]
    screen.blit(ship_img, (ship_x - ship_img.get_width() // 2, ship_y))
    screen.blit(font.render(f"Score: {score}   Time: {time_left}   Best:
{high_score}", True, WHITE), (10, 10))

    if game_over:
        screen.blit(big_font.render("GAME OVER", True, RED), (WIDTH // 2 - 160,
HEIGHT // 2 - 60))
        screen.blit(font.render(f"Final: {score}   Press R to restart", True,
WHITE), (WIDTH // 2 - 200, HEIGHT // 2 + 20))

    pygame.display.flip()
    clock.tick(60)

pygame.quit()

```

Play one round. What is different about firing M instead of SPACE, what do the sprites look like, and what does R do?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The sprite is supposed to show your `ship.png` picture, but the program crashes saying it cannot find the file.

```
ship_img = pygame.image.load("ship.png")
```

Hint: the picture file must be saved in the **same folder** as your code, with the exact name. Also use `.convert_alpha()` so the see-through background stays see-through.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The high score should remember your best across rounds, but it resets to 0 every restart.

```
if event.key == pygame.K_r and game_over:
    high_score = 0
    score = 0
    game_over = False
```

Hint: before resetting, compare the score to the best so far.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — After the timer hits 0, the player should not be able to keep flying or firing, but the game keeps playing.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]:
    ship_x -= SPEED
```

Hint: movement should only run while the game is not over.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Tune your two weapons

A missile is worth `e["points"] * 2` and travels slower. Change those numbers — make missiles faster, slower, or worth even more.

Write what you changed about the bullet and the missile:

Add a difficulty or a title screen

Pick one polish idea: a start screen before the game, or Easy/Hard modes that change the spawn speed or the timer.

Describe what you added and how it changes the game:

Hint: a title screen is just another `game_state` — show text and wait for a key before starting the timer.

5 · Make It

Finish your game and prepare your show & tell

Finish your polished game with two weapons, PNG sprites, a high score, and restart. Then get ready to present it.

Use these sentence starters for your presentation notes — write 4 to 6 sentences total.

My game is called ...

The function I am most proud of is ... because ...

My favourite polish feature is ...

The hardest bug I solved was ...

I fixed it by ...

The next thing I would add is ...

6 · Record Your Final Video 📹

Take a video of your finished game from start to game over to restart. Talk through it like you are presenting to the class. Try to use these words: **missile, sprite, image.load, high score, restart.**

1. Play a full round, firing both bullets and missiles, and let the timer run out.
2. Show the game-over screen and the high score.
3. Press R and show it restart.
4. Explain how you loaded your PNG sprites and one bug you fixed.

Write what you will say in your video. Plan it here before you record — this is your show & tell.

Submit ✓

Send this worksheet + your full-play video + your show & tell notes to teacher on KakaoTalk.